Linux Operating System

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2026/10/21

Today's Agenda

01 Why use Linux?

02 How to install Linux?

03 How to use Linux?

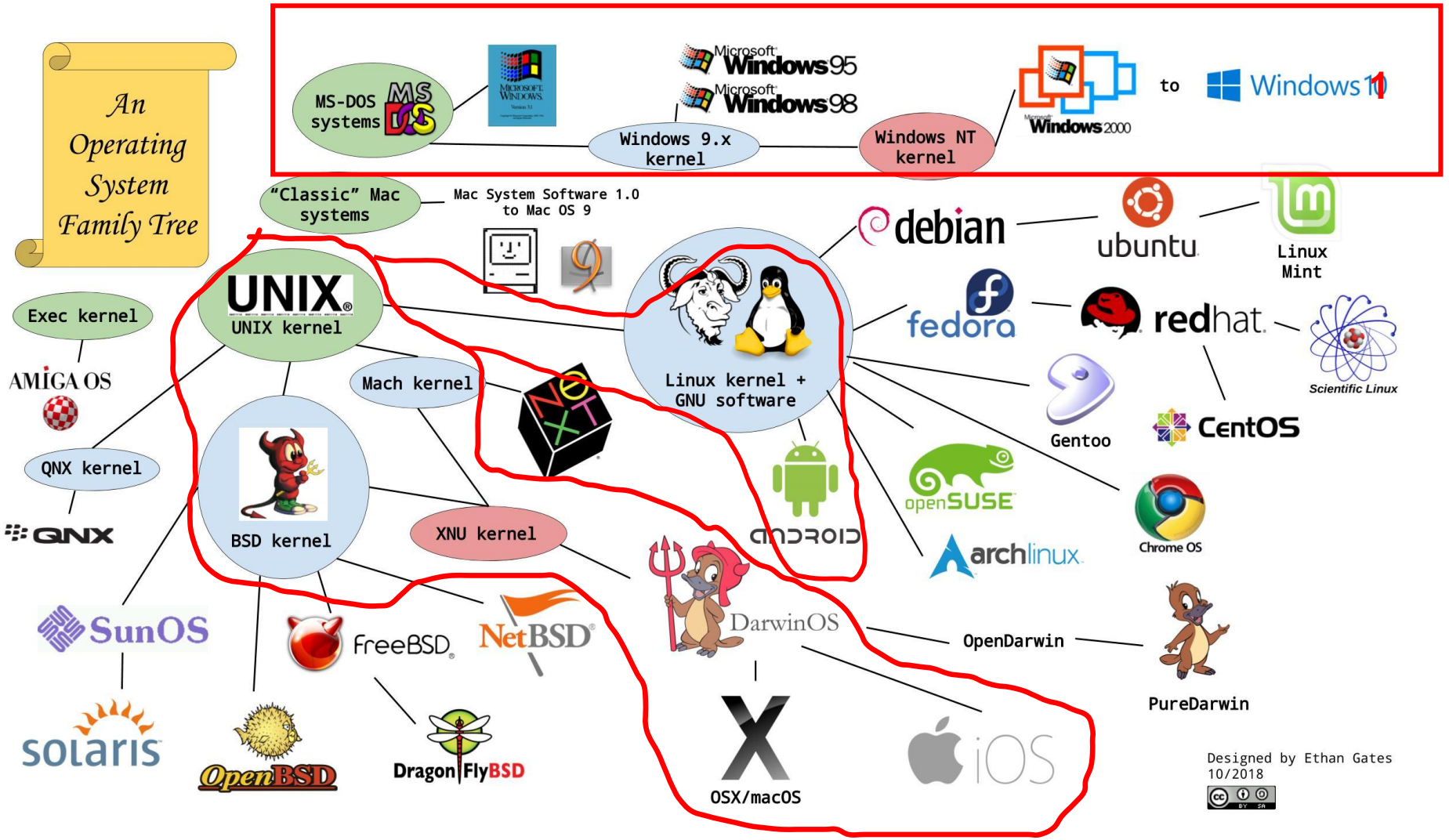
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Operating Systems



Operating Systems

Commercially successful operating systems



- **Advantages**
  - Easy to use
  - Many applications
  - Stable and reliable
- **Disadvantages**
  - Limited control over the system
  - Dependence on commercial ecosystems

Operating Systems

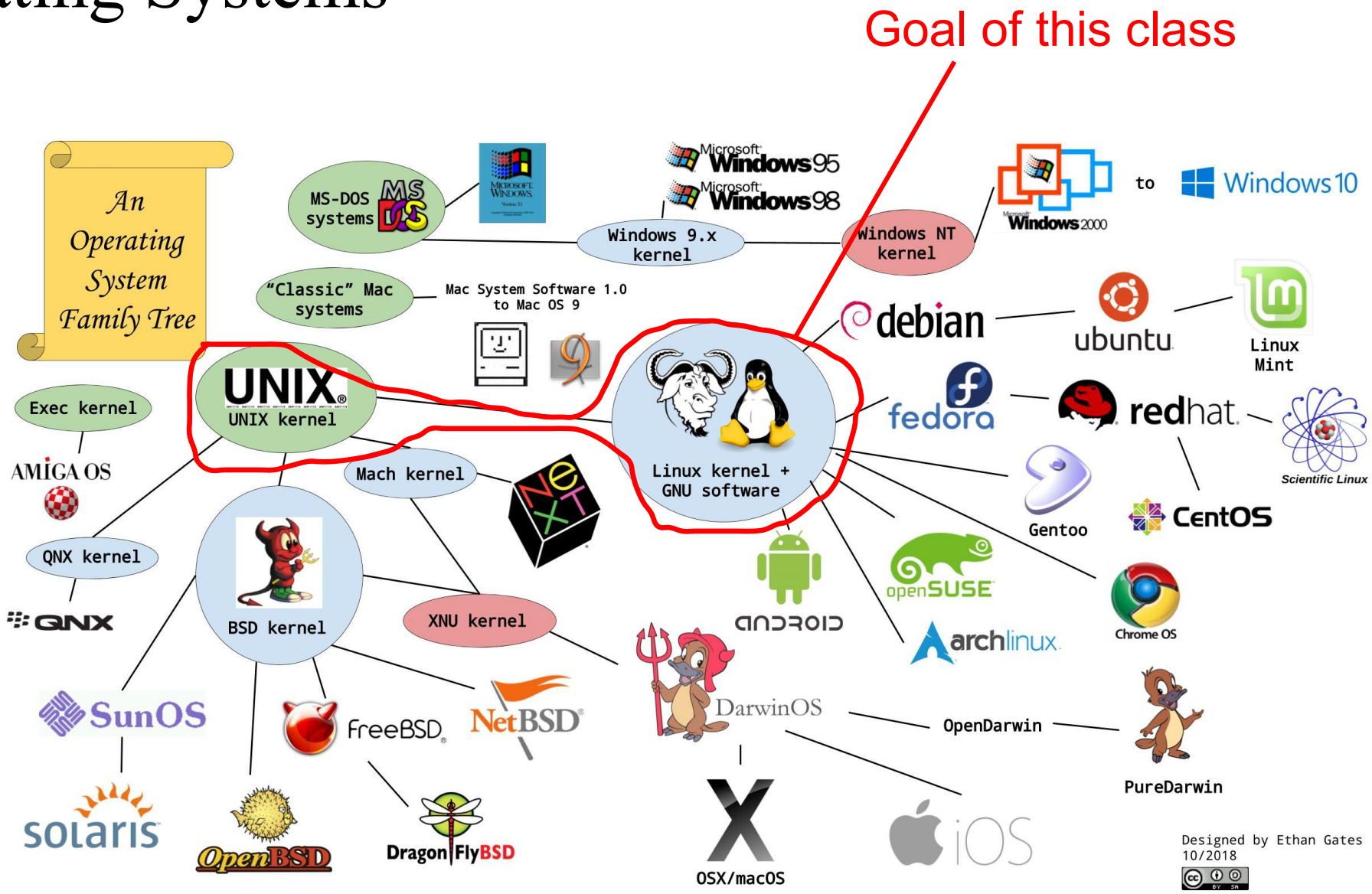


When you want to develop software or do research...

Linux is often a great choice because it is:

- **Free and open-source**
- **Flexible and customizable**
- **Widely used on servers and in research**

Operating Systems



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Installation (Windows)



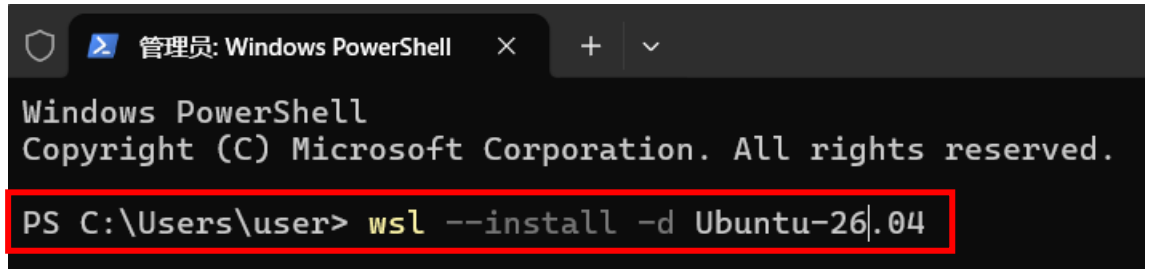
1) Right-click

2) Click

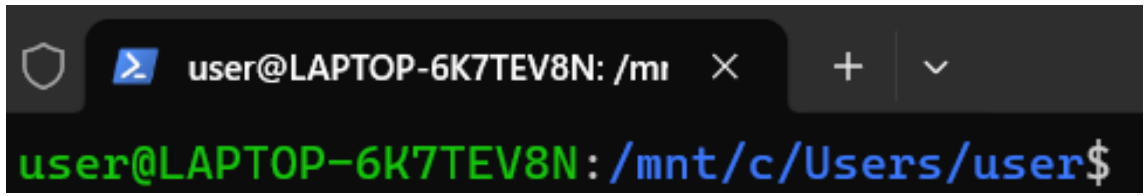


Installation (Windows)

3) Enter this command and press “Enter”, then follow its instructions.

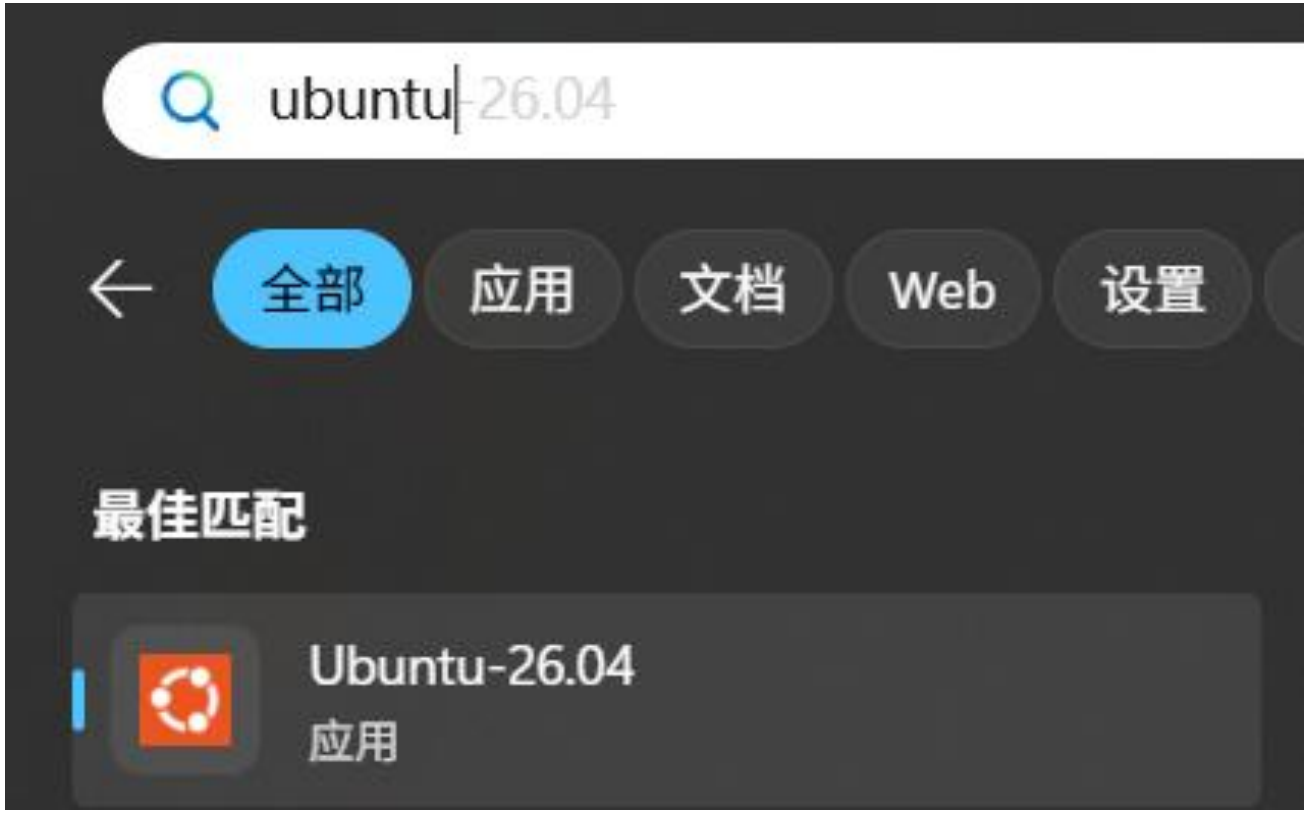


4) If you see sth like this, you succeed.



Installation (Windows)

After that, you can find it in your start menu.



Installation (MacOS)

MacOS is a Unix-based operating system, so you don't need to install Linux for this class. You can simply use the Terminal on your Mac.

Today's Agenda

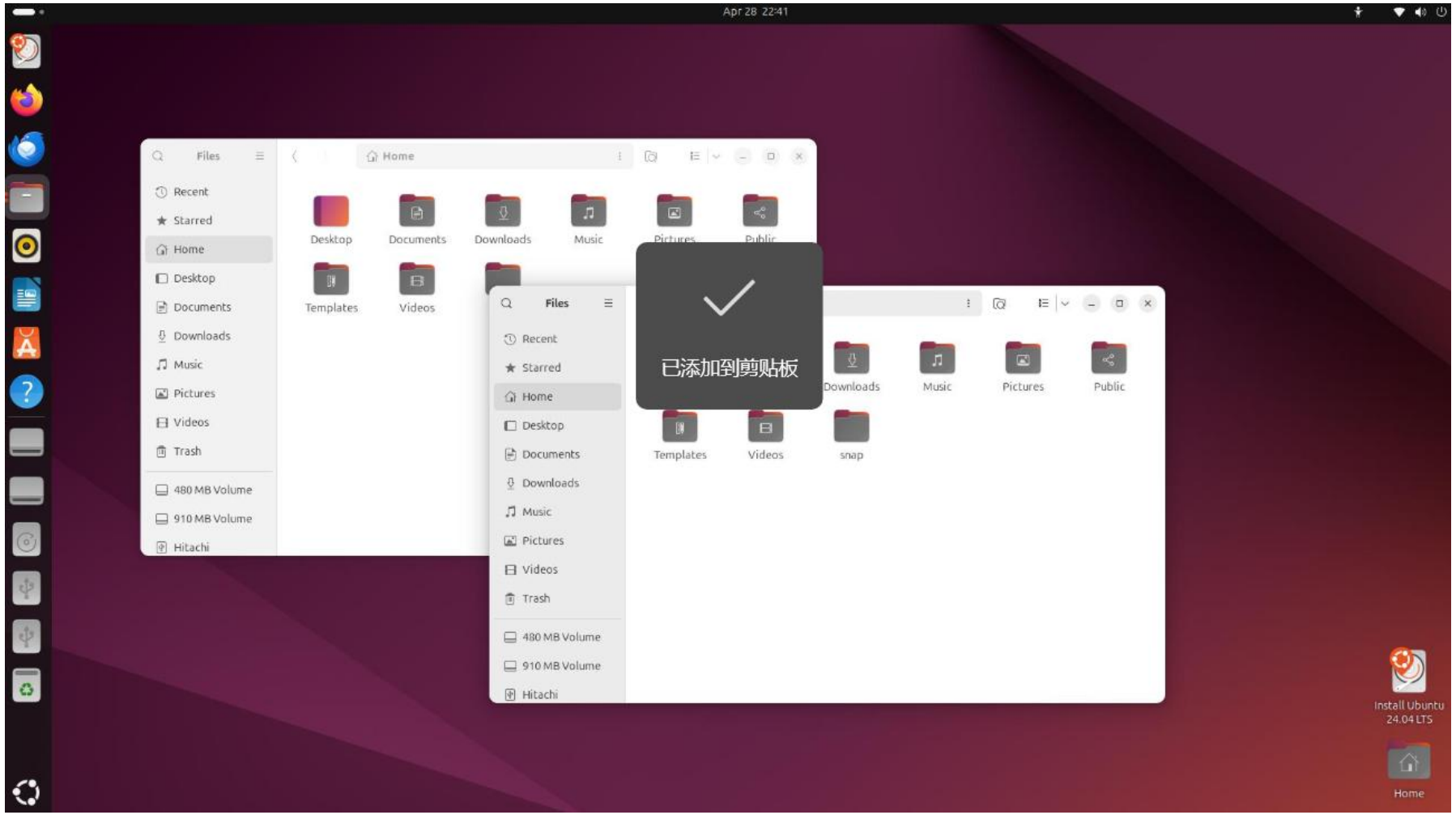
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GUI

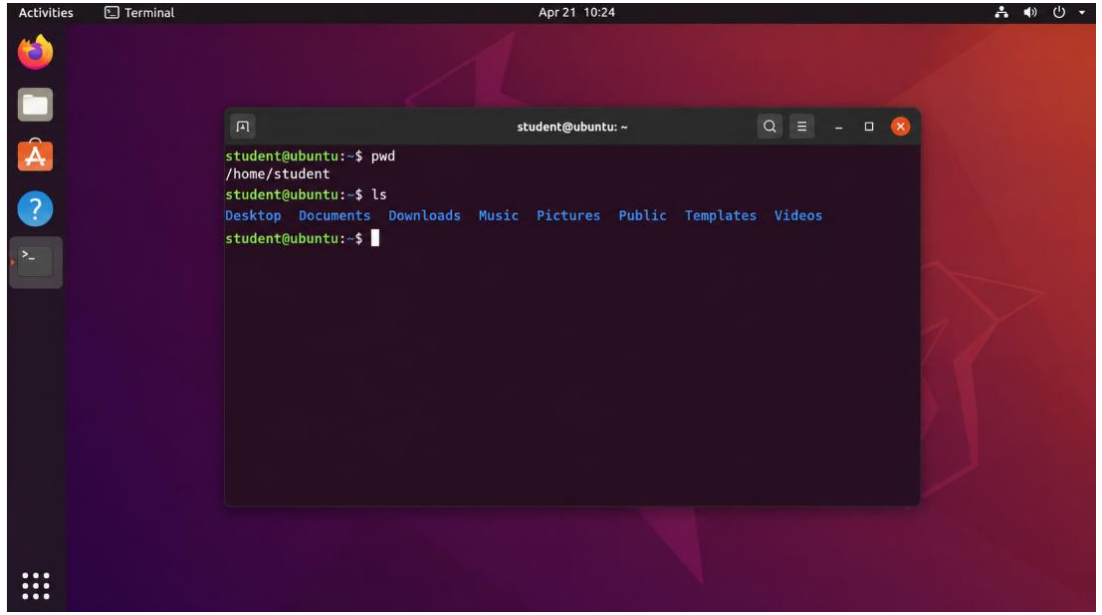
Most Linux distributions also provide a graphical user interface (GUI).



CLI

A command-line interface (CLI) allows users to interact with a computer by typing text commands instead of using a graphical interface.

CLI in Ubuntu system



CLI in WSL



GUI vs CLI

GUI — Graphical User Interface

Advantages
Easy to learn and visually intuitive.

Best for
Visual and interactive tasks.

Limitations
Repetitive tasks may require many clicks.

CLI — Command-Line Interface

Advantages
Fast, precise, and easy to automate.

Best for
Running programs and managing tools.

Limitations
Requires learning commands.

Reason to learn CLI

CLI and the File System

The CLI lets us navigate and manage files and directories using text commands.

Example file system

```
~  (/home/user)
├── source
│   └── v1
│       ├── Documents
│       └── Programs
└── configs
```

Common commands

`pwd`	Where am I?
`ls`	What is in this directory?
`cd Documents`	Go to another directory
`mkdir projects`	Create a directory
`cp / mv / rm`	Copy, move, or remove files

SAFETY!!!

A command operates relative to your current directory unless you specify another path.

Command-Line Safety

These commands can overwrite or permanently delete files.

`cp old.txt report.txt`	Overwrite an existing report.txt
`mv draft.txt report.txt`	Overwrite an existing report.txt
`rm -r project/`	Deletes the folder and everything inside it

Warning: In WSL, files under /mnt/ are real Windows files.

Files and Directories in Linux

In Linux, a directory is a special type of file.

Useful commands

`ls -lh`	List files and their sizes
`du -sh folder/`	Show the total disk usage of a folder
`stat file.txt`	Show detailed file information

Text File Editting

nano is a simple text editor you can use in the terminal.

Open (or create) a text file:

```
nano notes.txt
```

Edit the text, then use these shortcuts:

`Ctrl + O`	Save the file
`Enter`	Confirm the filename
`Ctrl + X`	Exit nano

Installing Software in Ubuntu

APT is Ubuntu's built-in package manager for installing, updating, and removing software.

1. Update the list of available software

```
sudo apt update
```

2. Install a software package (e.g., Git)

```
sudo apt install git
```

sudo = run the command with administrator privileges

Note: Ubuntu may ask for your password.

Where are the Softwares Installed?

Example: Git

`/usr/bin/git`	Executable program
`/usr/share/doc/git`	Documentation

Find the installed files:

`which git`	Find the executable
`dpkg -L git`	List files in the package

How to Use Installed Software

Each program defines its own commands and options.

Common pattern

```
program  [command]  [options]  [arguments]
```

Example: Git

`git status`	Check repository status
`git --help`	Show usage information

Running and Stopping Programs

A command can start a program that keeps running.

`sleep 30`	Waits for 30 seconds.
`Ctrl + C`	Interrupts the running program.

Try it: run sleep 30, then press Ctrl + C.

Checking and Stopping Processes

A process is a running instance of a program.

`ps aux`	Lists processes and their IDs (PIDs).
`kill 12345`	Requests that process 12345 stop.

PID = Process ID (a number identifying a process).

Only stop processes you recognize.

Checking Computer Resource Usage

Check which resources your computer is using.

`top`	Live process list and CPU / memory usage.
`free -h`	Shows memory usage.
`df -h`	Shows disk space usage.

Tip: Press q to exit top.